Project Proposal

MacroNews Portfolio Lab: A News- and Momentum-Driven Portfolio Backtesting Simulator for the U.S. Market

COMP3090: Course-related Project 2 | VinUniversity

Supervisor: Huynh Thanh Trung

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Full Name	Student ID	Role
Duong Hien Chi Kien	V202401461	Optimization Modeling
Cao Pham Minh Dang	V202401280	Data Engineer
Ngo Thanh An	V202401748	NLP Engineer
Nguyen Hoang Nam	V202401647	Backtesting & Evaluation
Tran Anh Chuong	V202401566	Full-stack Developer

Abstract

We propose a web-based simulator that backtests weekly portfolio allocation strategies on U.S. sector ETFs. Federal Reserve communications and financial news are processed by a multi-task FinBERT model into weekly topic-level sentiment, monetary-stance and sector-sentiment indices. These indices are mapped into *news views* on expected returns, while momentum, short-term reversal and volatility produce a separate set of *technical views*. The Black–Litterman model combines both view sets with market-implied equilibrium returns in a Bayesian manner, and a long-only mean-variance quadratic program with a turnover penalty determines the portfolio weights every week. A walk-forward backtest with ablation, placebo and bootstrap tests measures whether news adds information beyond price-based signals. All components are exposed through an interactive backtesting simulator (FastAPI, React, MySQL, Docker).

1 Problem Statement

Portfolio allocation requires two inputs: an estimate of expected returns and an estimate of risk. Risk (the covariance matrix) can be estimated reasonably well from historical prices, but expected returns cannot: sample means are extremely noisy, and the classical Markowitz optimizer amplifies this noise into extreme, unstable and poorly diversified portfolios. At the same time, a large and continuous stream of unstructured information—Federal Reserve statements, macroeconomic news and company news—carries information about future returns that is not directly usable by an optimizer.

The problem addressed by this project is therefore: *how can unstructured financial text be converted into quantitative, confidence-weighted return forecasts, combined with price-based signals, and turned into stable portfolio weights in a principled way—and does doing so actually improve out-of-sample portfolio performance?*

2 Motivation

- **Text matters for markets.** Monetary policy decisions, inflation surprises and labour-market news drive large rotations across U.S. sectors (e.g. rate-sensitive sectors such as utilities and real estate versus financials). Investors read this information, but rarely in a systematic, measurable way.
- **Markowitz is fragile.** Mean-variance optimization with sample-mean returns is known to underperform even the naive equally-weighted portfolio out of sample. A framework is needed that stays close to a sensible default and deviates only when there is credible information.
- **Black–Litterman is a natural bridge.** It treats each signal as a *view* with an explicit uncertainty and blends it with an equilibrium prior through Bayes' rule. This makes it possible to combine heterogeneous signals (text-based and price-based), each weighted by its own reliability.
- **A meaningful question.** Since prices already reflect much public information, the key question is not whether news "predicts the market", but whether news adds information *beyond* what momentum already captures. Our ablation design answers exactly this.
- **Course relevance.** The project integrates Natural Language Processing and Deep Learning (FinBERT fine-tuning), Machine Learning (regularized regression, walk-forward validation), Data Mining and Big Data (millions of news records), optimization and statistics (Bayesian estimation, convex optimization, bootstrap inference), and software engineering (a full-stack simulator).

3 Objectives and Research Questions

Objectives.

O1. Build a reproducible data pipeline for U.S. sector ETF prices, macroeconomic series, FOMC documents and large-scale financial news, stored in a relational database.

O2. Fine-tune a multi-task FinBERT model that classifies texts by macro topic and scores sentiment and hawkish/dovish stance, and aggregate the outputs into weekly news indices.

O3. Implement a Black–Litterman portfolio engine with two independent view sources (news and technical) and a convex mean-variance optimizer with realistic constraints, rebalanced weekly.

O4. Evaluate the system rigorously with a walk-forward backtest, ablation, placebo and statistical significance tests, free of look-ahead bias.

O5. Deliver an interactive web simulator that lets users configure, run, inspect and compare backtests.

Research questions.

- **RQ1.** How much does fine-tuned FinBERT improve topic, sentiment and stance classification over lexicon-based and TF-IDF baselines?
- **RQ2.** Do news views improve risk-adjusted returns or reduce drawdowns compared with no views, technical views only, and standard allocation baselines? In particular, does *news + technical* outperform *technical only*?
- **RQ3.** How do the strategies behave during stress periods (2020 COVID crash, 2022 rate-hike cycle)?

4 Methodology

4.1 System Overview

The system is split into an **offline pipeline**, executed once over the full history (and re-run only when data or models change), and an **online simulator**, which runs a walk-forward backtest on demand. Figure 1 shows the data flow.

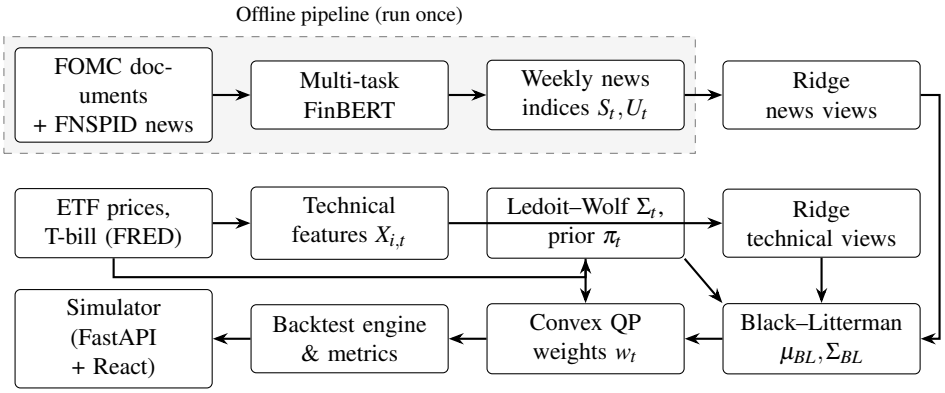


Figure 1: Data flow. The upper-left block (text processing) is computed offline and stored in MySQL; everything else is recomputed every week inside the walk-forward backtest.

4.2 Data

Investment universe. The core universe contains $N = 11$ assets: the nine original SPDR sector ETFs (XLB, XLE, XLF, XLI, XLK, XLP, XLU, XLV, XLY), a long-term Treasury ETF (TLT) and a gold ETF (GLD). XLRE (from 2015) and XLC (from 2018) are available as optional assets in the simulator from their inception dates. SPY is used only as a benchmark. Using sector ETFs instead of individual stocks keeps the covariance matrix well-conditioned and avoids company-level entity linking.

Sources.

- *Prices:* daily adjusted close prices from Yahoo Finance (`yfinance`), 2010–2023.
- *Risk-free rate and macro series:* 3-month T-bill rate (DTB3) and selected macro series from FRED (`fredapi`).
- *Federal Reserve communications:* FOMC statements, minutes and press-conference transcripts from the Federal Reserve website.
- *Financial news:* FNSPID [9], a public dataset of 15.7 million time-stamped financial news records linked to S&P 500 tickers (1999–2023). Each record contains date, title, ticker, publisher and article text.
- *Labelled NLP data:* Financial PhraseBank [7] (sentiment), the FOMC hawkish/dovish dataset of Shah et al. [8] (stance), and our own manually labelled set for macro topics.

Storage. All data are stored in MySQL with the main tables `assets`, `prices_daily`, `documents`, `doc_scores`, `news_index_weekly` and `backtest_runs`. Each score row records the model version, so that different NLP models can be compared in the simulator.

4.3 Text Processing (Offline)

Step 1 — Cleaning and alignment. Texts are normalized (HTML removal, Unicode normalization), near-duplicates are removed with MinHash similarity [12] (threshold 0.85), and every timestamp is converted to U.S. Eastern time. A document published after the Friday 16:00 ET cut-off belongs to the following week. FNSPID tickers are mapped to their GICS sector, which links company news to a sector ETF.

Step 2 — Units of analysis. For news, the input is the title plus the first paragraph (max. 256 tokens). Long FOMC documents are split into sentences, each sentence is classified, and document scores are averages over sentences classified as relevant.

Step 3 — Multi-task FinBERT. A shared FinBERT encoder [6] produces the `[CLS]` embedding $\boldsymbol{h}_j \in \mathbb{R}^{768}$ for text j, followed by three linear softmax heads:

$$\begin{aligned}
\boldsymbol{p}_j^{\text{topic}} &= \text{softmax}(W_T \boldsymbol{h}_j + \boldsymbol{b}_T) \in \Delta^7, && \{\text{monetary, inflation, labour, growth, energy, global, other}\} \\
\boldsymbol{p}_j^{\text{sent}} &= \text{softmax}(W_S \boldsymbol{h}_j + \boldsymbol{b}_S) \in \Delta^3, && \{\text{negative, neutral, positive}\} \\
\boldsymbol{p}_j^{\text{stance}} &= \text{softmax}(W_H \boldsymbol{h}_j + \boldsymbol{b}_H) \in \Delta^3, && \{\text{dovish, neutral, hawkish}\}
\end{aligned}$$

Training minimizes $\mathscr{L} = \mathscr{L}_{\text{topic}} + \lambda_S \mathscr{L}_{\text{sent}} + \lambda_H \mathscr{L}_{\text{stance}}$ (cross-entropy), where each loss term is masked when a training example has no label for that task, so the three datasets can be combined. About 1,500 documents are manually labelled for topic by the team following a written guideline; 200 of them are double-annotated to report Cohen's κ.

Step 4 — Output of the NLP stage. Each document is stored as one row of `doc_scores`:

```
(doc_id, timestamp, sector, p_topic[7], p_sent[3], p_stance[3], model_version)
```

from which we derive a polarity score $s_j = p_{j,\text{pos}}^{\text{sent}} - p_{j,\text{neg}}^{\text{sent}} \in [-1,1]$ and a stance score $\eta_j = p_{j,\text{hawk}}^{\text{stance}} - p_{j,\text{dove}}^{\text{stance}} \in [-1,1]$.

4.4 Weekly News Indices (Offline)

Let t denote the Friday 16:00 ET decision time of week t and τ_j the timestamp of document j. Documents decay exponentially with half-life $H = 14$ days, $\omega_j(t) = \exp\big(-\ln 2\,(t-\tau_j)/H\big)$, using documents from the last eight weeks only. For each macro topic $k \in \{1,\dots,6\}$:

$$S_t^{(k)} = \frac{\sum_{j:\,\tau_j \le t} \omega_j(t)\, p_{j,k}^{\text{topic}}\, s_j}{\sum_{j:\,\tau_j \le t} \omega_j(t)\, p_{j,k}^{\text{topic}}}, \qquad H_t = \frac{\sum_j \omega_j(t)\, p_{j,\text{mon}}^{\text{topic}}\, \eta_j}{\sum_j \omega_j(t)\, p_{j,\text{mon}}^{\text{topic}}}.$$

A sector sentiment index $S_{g,t}^{\text{sec}}$ is computed in the same way over company news whose ticker belongs to sector g. A news uncertainty index U_t is the weighted cross-sectional standard deviation of s_j in week t combined with abnormal news volume. Every index is standardized with a rolling 156-week mean and standard deviation computed from *past values only*. The resulting feature vector for asset i is

$$\boldsymbol{F}_{i,t}^{\text{news}} = \big[S_t^{(1)}, \dots, S_t^{(6)},\, H_t,\, S_{g(i),t}^{\text{sec}}\big]^\top \in \mathbb{R}^8,$$

where the sector term is omitted for TLT and GLD.

4.5 Technical Features

Technical indicators enter the model *only* as a second, independent source of views. For each asset i with weekly excess returns $r_{i,t}$:

- **Momentum:** $\mathrm{MOM}_{i,t} = \prod_{s=t-11}^{t-1}(1+r_{i,s}) - 1$ (12-week return, skipping the latest week);
- **Short-term reversal:** $\mathrm{REV}_{i,t} = r_{i,t}$ (latest weekly return);
- **Volatility:** $\mathrm{VOL}_{i,t}$ = annualized realized volatility over the last 20 trading days.

Each feature is standardized cross-sectionally across assets, giving $\boldsymbol{X}_{i,t} \in \mathbb{R}^3$. Other common indicators (RSI, MACD) are largely transformations of momentum and reversal and are deliberately excluded to limit overfitting.

4.6 Rebalancing Protocol

The portfolio is rebalanced **once per week**. At the Friday close t the information set contains only documents with $\tau_j \le t$ and prices up to that close. Target weights are computed over the weekend and executed at the **next trading day's close** (a one-day implementation lag, which is conservative). Between rebalancing dates the portfolio is marked to market daily and weights drift with prices. Forecasts use a horizon of $h = 4$ weeks (about 20 trading days); weekly rebalancing with a 4-week horizon is standard and lets the portfolio react to new information without chasing one-week noise. Weekly versus monthly rebalancing is compared in the sensitivity analysis.

4.7 Risk Model and Equilibrium Prior

Covariance. Let $\hat{S}_t$ be the sample covariance of daily excess returns over the last 252 trading days. The Ledoit–Wolf estimator [4] shrinks it toward a scaled identity,

$$\hat{\Sigma}_t^{LW} = (1-\rho^*)\,\hat{S}_t + \rho^*\,\nu I, \qquad \nu = \mathrm{tr}(\hat{S}_t)/N,$$

with the analytically optimal intensity ρ^*. It is scaled to the forecast horizon: $\Sigma_t = 20\,\hat{\Sigma}_t^{LW}$.

Prior (reverse optimization). An unconstrained mean-variance investor with risk aversion δ maximizes $w^\top \mu - \frac{\delta}{2} w^\top \Sigma w$. The first-order condition $\mu - \delta \Sigma w = 0$ states that the optimal portfolio satisfies $\mu = \delta \Sigma w$. Assuming the reference portfolio w_{ref} is optimal, the implied equilibrium excess returns are

$$\pi_t = \delta\,\Sigma_t\, w_{\mathrm{ref}},$$

with $\delta = 2.5$. The reference portfolio allocates 70% to equities (split across sectors by S&P 500 sector weights, or equally if historical weights are unavailable), 20% to TLT and 10% to GLD. The prior is modelled as $\mu \sim \mathcal{N}(\pi_t, \tau\Sigma_t)$ with $\tau = 0.05$.

4.8 View Generation

For each asset i and each view source $v \in \{\mathrm{news}, \mathrm{tech}\}$ we fit a Ridge regression that forecasts the forward 4-week excess return $y_{i,s} = \sum_{u=s+1}^{s+4} r_{i,u}$:

$$(\hat{a}_i, \hat{\boldsymbol{b}}_i) = \arg\min_{a,\boldsymbol{b}} \sum_{s \in \mathscr{T}_t} \left(y_{i,s} - a - \boldsymbol{b}^\top \boldsymbol{Z}_{i,s}\right)^2 + \alpha \|\boldsymbol{b}\|_2^2, \qquad \boldsymbol{Z} = \boldsymbol{F}^{\mathrm{news}} \text{ or } \boldsymbol{X}.$$

No look-ahead: the training set is $\mathscr{T}_t = \{s : s+4 \le t\}$, i.e. only weeks whose 4-week target is fully realized at time t (purging of overlapping targets). Models are re-fitted every week on an expanding window (minimum 156 weeks). The penalty α is selected by time-series cross-validation with a 4-week gap inside the validation period and then fixed.

A view is created for asset i only if the signal is sufficiently strong, $|\hat{q}_i| > 0.5\,\hat{\sigma}_i$, where

$$\hat{q}_{i,t}^{v} = \hat{a}_i + \hat{\boldsymbol{b}}_i^\top \boldsymbol{Z}_{i,t}, \qquad \Omega_{ii}^{\mathrm{news}} = c_{\mathrm{news}}\,\hat{\sigma}_{\mathrm{news},i}^2\,(1+\kappa U_t^+), \qquad \Omega_{ii}^{\mathrm{tech}} = c_{\mathrm{tech}}\,\hat{\sigma}_{\mathrm{tech},i}^2,$$

and $\hat{\sigma}^2$ is the out-of-fold residual variance of the regression. A model that forecasts poorly therefore produces views with low confidence, and news views are automatically down-weighted when news is contradictory (U_t high). Each view is an absolute view: its row of the pick matrix is the unit vector $e_i^\top$. The two sources are stacked:

$$P = \begin{bmatrix} P^{\text{news}} \\ P^{\text{tech}} \end{bmatrix} \in \mathbb{R}^{K \times N}, \quad q = \begin{bmatrix} q^{\text{news}} \\ q^{\text{tech}} \end{bmatrix} \in \mathbb{R}^{K}, \quad \Omega = \begin{bmatrix} \Omega^{\text{news}} & 0 \\ 0 & \Omega^{\text{tech}} \end{bmatrix}.$$

An asset may receive both a news view and a technical view; the block-diagonal Ω assumes that the errors of the two sources are independent, a simplification we state explicitly.

4.9 Black–Litterman Posterior

The views state $P\mu = q + \varepsilon$ with $\varepsilon \sim \mathcal{N}(0, \Omega)$. Combining prior and views by Bayes' rule, the posterior mean is the maximum a posteriori estimate, i.e. the solution of the generalized least-squares problem [1, 2]

$$\mu_{BL} = \arg\min_{\mu} \; (\mu - \pi)^\top (\tau\Sigma)^{-1} (\mu - \pi) + (P\mu - q)^\top \Omega^{-1} (P\mu - q).$$

Setting the gradient to zero, $(\tau\Sigma)^{-1}(\mu - \pi) + P^\top \Omega^{-1}(P\mu - q) = 0$, gives

$$\mu_{BL} = M\big[(\tau\Sigma)^{-1}\pi + P^\top \Omega^{-1} q\big], \qquad M = \big[(\tau\Sigma)^{-1} + P^\top \Omega^{-1} P\big]^{-1}, \qquad \Sigma_{BL} = \Sigma + M.$$

Numerically we use the equivalent Woodbury form $\mu_{BL} = \pi + \tau\Sigma P^\top \big(P\tau\Sigma P^\top + \Omega\big)^{-1}(q - P\pi)$, which only inverts a $K \times K$ matrix and makes the intuition explicit: the prior is corrected by the *surprise* $q - P\pi$, weighted by the relative precision of views and prior. With no views ($K = 0$), $\mu_{BL} = \pi$ and the optimizer returns (approximately) w_{ref}.

4.10 Portfolio Optimization

Formulation. Let $\tilde{w}_t$ be the drifted weights just before rebalancing. The target weights solve

$$\boxed{\max_{w \in \mathbb{R}^N} \; w^\top \mu_{BL} - \frac{\gamma}{2} w^\top \Sigma_{BL} w - \lambda \, \|w - \tilde{w}_t\|_1 \quad \text{s.t.} \quad \mathbf{1}^\top w = 1, \;\; 0 \le w_i \le w_{\max}}$$

with $\gamma = \delta = 2.5$ (consistent with the prior), $w_{\max} = 0.30$ and turnover penalty λ (e.g. 10 bps). The three terms are expected return, risk (variance including all correlations) and an approximation of transaction costs. Short selling and leverage are not allowed.

Closed form via Lagrange multipliers (budget constraint only). Ignoring the bounds and the turnover term, the Lagrangian is

$$\mathcal{L}(w, \nu) = w^\top \mu_{BL} - \tfrac{\gamma}{2} w^\top \Sigma_{BL} w - \nu(\mathbf{1}^\top w - 1).$$

Stationarity gives $\mu_{BL} - \gamma\Sigma_{BL} w - \nu\mathbf{1} = 0$, so $w^* = \frac{1}{\gamma}\Sigma_{BL}^{-1}(\mu_{BL} - \nu\mathbf{1})$, and the budget constraint yields

$$\nu^* = \frac{\mathbf{1}^\top \Sigma_{BL}^{-1} \mu_{BL} - \gamma}{\mathbf{1}^\top \Sigma_{BL}^{-1} \mathbf{1}}.$$

The multiplier ν^* is the marginal utility of one extra unit of capital. This closed form is used as a unit test for the numerical solver and to explain the role of Σ^{-1}: weights are expected returns adjusted for risk *and* correlation.

Full problem: KKT conditions. With multipliers ν for the budget, $\underline{\beta}_i \ge 0$ for $w_i \ge 0$ and $\overline{\beta}_i \ge 0$ for $w_i \le w_{\max}$, an optimal w^* satisfies

$$\mu_{BL} - \gamma\Sigma_{BL} w^* - \lambda g - \nu\mathbf{1} + \underline{\boldsymbol{\beta}} - \overline{\boldsymbol{\beta}} = 0, \quad g \in \partial\|w^* - \tilde{w}_t\|_1, \quad \underline{\beta}_i w_i^* = 0, \quad \overline{\beta}_i (w_{\max} - w_i^*) = 0.$$

Interpretation: an asset is held at zero weight when its risk-adjusted marginal return is below the marginal value of capital ν, and the subgradient g creates a "no-trade zone" in which small changes in views do not trigger trades.

Solving. The ℓ_1 term is linearized with auxiliary variables $u \in \mathbb{R}^N$: $u \geq w - \tilde{w}_t$, $u \geq \tilde{w}_t - w$, and the penalty becomes $\lambda \mathbf{1}^\top u$. The result is a convex quadratic program (concave objective since $\Sigma_{BL} \succ 0$, linear constraints), so it has a unique global optimum. It is solved with CVXPY [16] using the Clarabel interior-point solver (OSQP [17] as fallback); an $N = 11$ problem solves in milliseconds, so a 10-year weekly backtest takes a few seconds.

Baseline strategies use the same constraints: equal weight ($w_i = 1/N$) [5]; minimum variance ($\min w^\top \Sigma w$); inverse volatility ($w_i \propto 1/\sigma_i$); Markowitz with 156-week sample-mean returns; Black–Litterman with no views; and SPY buy-and-hold.

4.11 Walk-forward Backtest

Algorithm 1 Weekly walk-forward backtest (online simulator)

Require: period $[T_0, T_1]$, asset set, strategy, parameters $(\delta, \tau, \gamma, w_{\max}, \lambda, c)$

$w \leftarrow w_{\text{ref}}$, $V \leftarrow 1$
for each Friday $t \in [T_0, T_1]$ **do**
  Load S_t, U_t and prices up to t from MySQL ▷ precomputed offline
  $\Sigma_t \leftarrow$ Ledoit–Wolf on last 252 days; $\pi_t \leftarrow \delta \Sigma_t w_{\text{ref}}$
  Fit Ridge models on $\{s : s+4 \leq t\}$; build P, q, Ω for the enabled view sources
  $(\mu_{BL}, \Sigma_{BL}) \leftarrow \text{BlackLitterman}(\Sigma_t, \pi_t, P, q, \Omega, \tau)$
  $w^* \leftarrow \text{SolveQP}(\mu_{BL}, \Sigma_{BL}, \tilde{w}_t)$
  Execute at next close: $V \leftarrow V\,(1 - c\,\|w^* - \tilde{w}_t\|_1)$
  Mark to market daily until the next rebalance; record weights, views and trades
end for
Compute metrics and store the run in `backtest_runs`

Transaction costs are $c = 5$ bps per unit of one-way turnover (sector ETFs are highly liquid). The data are split chronologically: 2010–2012 warm-up, **2013–2017 validation** (all hyperparameters: α, c_{news}, c_{tech}, κ, λ), and **2018–2023 out-of-sample test**, which is touched only once for the final results and contains the 2020 and 2022 stress periods.

4.12 Evaluation

NLP (RQ1). Macro-F1 per task on a held-out test set for: (a) Loughran–McDonald lexicon [10], (b) TF-IDF + logistic regression, (c) pre-trained FinBERT without fine-tuning, (d) our multi-task fine-tuned FinBERT. Inter-annotator agreement is reported with Cohen's κ.

Portfolio (RQ2, RQ3). Annualized return and volatility, Sharpe and Sortino ratios, maximum drawdown, weekly $\text{CVaR}_{95\%}$ [13], and average turnover. The ablation compares Black–Litterman with (i) no views, (ii) technical views only, (iii) news views only, (iv) both. Additional checks:

- **Significance:** Sharpe-ratio differences are tested with the stationary bootstrap [14] (mean block length 8 weeks, 5,000 resamples), following Ledoit and Wolf [15].
- **Placebo:** news indices are circularly shifted by a random offset of at least 52 weeks (preserving their autocorrelation) and the backtest is repeated 200 times; if real news is not better than most placebo runs, the gain is attributed to chance.
- **NLP quality to portfolio:** the news strategy is re-run with lexicon and TF-IDF scores to test whether better classification translates into better portfolios.

- **Sensitivity:** τ, γ, $w_{\max}$, half-life H, and weekly versus monthly rebalancing.

4.13 Simulator

The simulator only replays *historical* data; it performs no live or paper trading. It offers:

- **Configuration:** period, assets, strategy, view sources (news / technical / both), NLP model version, δ, τ, γ, $w_{\max}$, λ, costs, rebalancing frequency.
- **Results:** equity curves versus SPY and equal weight, drawdown chart, stacked weight chart over time, metrics table.
- **Comparison:** overlay several saved runs (e.g. the ablation strategies, or different parameters and periods) on the same charts and table.
- **Weekly drill-down:** for any rebalancing date, show the news indices, the most influential articles, all views (q, Ω), the prior and posterior returns, and the weights before and after rebalancing, which makes every decision explainable.

Architecture. MySQL (data and saved runs), FastAPI (endpoints such as `POST /backtest`, `GET /runs`, `GET /compare`, `GET /week/{date}`), and a React + TypeScript front end (Tailwind CSS, shadcn/ui, TradingView Lightweight Charts, Recharts). Everything is packaged with Docker Compose (services `db`, `backend`, `frontend`, and an on-demand `pipeline`). FinBERT fine-tuning runs on a cloud GPU; the simulator itself needs no GPU because all text scores are precomputed.

5 Expected Outputs

1. A cleaned database of prices, FOMC documents and financial news, with document-level scores and weekly news indices, plus a reproducible ingestion pipeline.
2. A manually labelled macro-topic dataset with annotation guideline and agreement statistics.
3. A fine-tuned multi-task FinBERT model and its comparison with baselines.
4. A tested Python portfolio engine (Black–Litterman with two view sources, convex optimizer, walk-forward backtester).
5. A Dockerized web simulator for configuring, running, inspecting and comparing backtests.
6. Progress report, final presentation and final report with all evaluation results.

6 Timeline

Week	Dates	Tasks
3	Oct 5–9	Proposal and supervisor approval; project description submitted; repository and Docker skeleton
4–5	Oct 12–23	Price, FRED and FOMC ingestion; FNSPID download, filtering and deduplication; MySQL schema; labelling guideline; first BL and optimizer module with unit tests; UI wireframes
6–7	Oct 26–Nov 6	Manual labelling and Cohen's κ; lexicon and TF-IDF baselines; technical views; baseline strategies backtested; FastAPI skeleton
8	Nov 9–13	**Progress report** (Nov 13)
9–11	Nov 16–Dec 4	Multi-task FinBERT fine-tuning; scoring the corpus; weekly news indices; news views; full Black–Litterman integration
12–13	Dec 7–18	Full backtests on validation and test periods; ablation, placebo, bootstrap and sensitivity analyses
14–15	Dec 21–Jan 1	Simulator completion (comparison and drill-down views); Dockerization; supervisor feedback; report drafting
16	Jan 4–8	**Final presentation** (Jan 8)
17	Jan 11–15	**Final report** (Jan 15)

Responsibilities. Data pipeline and database: Cao Pham Minh Dang. NLP labelling and FinBERT: Ngo Thanh An (all members annotate). Risk model, Black–Litterman and optimizer: Duong Hien Chi Kien. Backtest engine and statistical evaluation: Nguyen Hoang Nam. API, front end and Docker: Tran Anh Chuong.

7 Risks and Mitigation

Risk	Mitigation
News signal is weak or absent	A negative result is still valid; the contribution is the rigorous measurement (ablation, placebo, bootstrap), not alpha
Uneven FNSPID coverage across years	Report coverage statistics; normalize indices by rolling statistics; use FOMC texts as a stable macro source
Overfitting of hyperparameters	Strict validation/test split; test set used once; small feature sets with Ridge regularization
Look-ahead bias	Timestamp cut-off, purged training windows, one-day execution lag, unit tests on data availability
Schedule slippage	Technical views and baselines are completed first so a working system exists by the progress report

Ethics. All data are public and used under their terms of use and licenses. The simulator is an educational research tool and does not constitute investment advice; results are historical and include explicit cost assumptions.

References

[1] F. Black and R. Litterman. Global portfolio optimization. *Financial Analysts Journal*, 48(5):28–43, 1992.

[2] G. He and R. Litterman. The intuition behind Black–Litterman model portfolios. Goldman Sachs Investment Management Research, 1999.

[3] H. Markowitz. Portfolio selection. *The Journal of Finance*, 7(1):77–91, 1952.

[4] O. Ledoit and M. Wolf. A well-conditioned estimator for large-dimensional covariance matrices. *Journal of Multivariate Analysis*, 88(2):365–411, 2004.

[5] V. DeMiguel, L. Garlappi, and R. Uppal. Optimal versus naive diversification: How inefficient is the 1/N portfolio strategy? *The Review of Financial Studies*, 22(5):1915–1953, 2009.

[6] D. Araci. FinBERT: Financial sentiment analysis with pre-trained language models. arXiv:1908.10063, 2019.

[7] P. Malo, A. Sinha, P. Korhonen, J. Wallenius, and P. Takala. Good debt or bad debt: Detecting semantic orientations in economic texts. *Journal of the Association for Information Science and Technology*, 65(4):782–796, 2014.

[8] A. Shah, S. Paturi, and S. Chava. Trillion dollar words: A new financial dataset, task & market analysis. In *Proceedings of ACL*, 2023.

[9] Z. Dong, X. Fan, and Z. Peng. FNSPID: A comprehensive financial news dataset in time series. arXiv:2402.06698, 2024.

[10] T. Loughran and B. McDonald. When is a liability not a liability? Textual analysis, dictionaries, and 10-Ks. *The Journal of Finance*, 66(1):35–65, 2011.

[11] N. Jegadeesh and S. Titman. Returns to buying winners and selling losers: Implications for stock market efficiency. *The Journal of Finance*, 48(1):65–91, 1993.

[12] A. Z. Broder. On the resemblance and containment of documents. In *Proceedings of Compression and Complexity of Sequences*, 1997.

[13] R. T. Rockafellar and S. Uryasev. Optimization of conditional value-at-risk. *Journal of Risk*, 2(3):21–41, 2000.

[14] D. N. Politis and J. P. Romano. The stationary bootstrap. *Journal of the American Statistical Association*, 89(428):1303–1313, 1994.

[15] O. Ledoit and M. Wolf. Robust performance hypothesis testing with the Sharpe ratio. *Journal of Empirical Finance*, 15(5):850–859, 2008.

[16] S. Diamond and S. Boyd. CVXPY: A Python-embedded modeling language for convex optimization. *Journal of Machine Learning Research*, 17(83):1–5, 2016.

[17] B. Stellato, G. Banjac, P. Goulart, A. Bemporad, and S. Boyd. OSQP: An operator splitting solver for quadratic programs. *Mathematical Programming Computation*, 12(4):637–672, 2020.